ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Paper 1 style (theory). Section A: objective questions (30) · Section B: what robots are, history and types, the sense-think-act cycle, sensors, motors and gears, degrees of freedom, configurations and wheel speed, safety, ethics, maintenance and cybersecurity (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A robot is best described as a machine that can …

A sense, think and act on its own. B only move.
C only show pictures. D only store data.

1.1.2 Robots that weld and paint car bodies in a factory are …

A service robots. B industrial robots. C toy robots. D space robots.

1.1.3 The "brain" of a robot is its …

A sensor. B actuator. C controller. D chassis.

1.1.4 The sensor most often used to follow a black line on the floor is a/an …

A microphone. B thermometer.
C GPS receiver. D infrared reflective sensor.

1.1.5 The motor that turns to a set angle and holds it is a …

A DC motor. B servo motor. C fan motor. D generator.

1.1.6 The movement of a wheel turning on its axle is …

A linear. B reciprocating. C rotary. D oscillating.

1.1.7 A robot that can only move forwards and backwards along a track has … degree(s) of freedom.

A 1 B 2 C 3 D 6

1.1.8 A robot's touch (bump) sensor is usually a simple …

A switch. B LED. C capacitor. D speaker.

1.1.9 A battery stores energy as …

A sound energy. B chemical energy. C nuclear energy. D light energy.

1.1.10 3D printing is a/an … manufacturing process.

A subtractive B casting C forging D additive

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The word "robot" was introduced by the Czech writer Karel Čapek.

1.2.2 A robot vacuum cleaner is an example of a service robot.

1.2.3 Sensors are the outputs of a robot.

1.2.4 An ultrasonic sensor uses sound waves to measure distance.

1.2.5 The direction of a DC motor can be reversed by swapping its supply wires.

1.2.6 Robots never need maintenance.

1.2.7 A gripper is a type of end effector.

1.2.8 Robots can only be programmed in one programming language.

1.2.9 An emergency-stop button must be easy to reach.

1.2.10 Wheeled robots climb stairs more easily than legged robots.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Body or frame of a mobile robot	A	microcontroller
1.3.2	Part that makes a robot move, such as a motor	B	program
1.3.3	Sensor whose resistance changes with light	C	chassis
1.3.4	Tool on the end of a robot arm	D	3D printing
1.3.5	Set of instructions that a robot follows	E	end effector
1.3.6	Robot controlled by a person from a distance	F	legged robot
1.3.7	Small computer on one chip that controls a robot	G	actuator
1.3.8	All the space a robot arm can reach	H	work envelope
1.3.9	Making an object layer by layer from a computer model	I	LDR
1.3.10	Robot that walks	J	teleoperated robot

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: ROBOTS, HISTORY AND TYPES

2.1 Define the term robot. (2)

2.2 Name THREE important people or events in the history of robots. (3)

2.3 Describe FOUR types of mobile robots according to how they move. (4)

2.4 Give FOUR reasons why companies use robots. (4)

2.5 Robots are often used for the "three Ds". Name them and give ONE example job for EACH. (3)

[16]

QUESTION 3: SENSORS AND ACTUATORS

3.1 Explain the sense–think–act cycle, using a robot vacuum cleaner as an example. (3)

3.2 For EACH sensor, explain briefly how it works and give ONE use on a robot: (a) touch switch; (b) infrared reflective sensor; (c) ultrasonic sensor; (d) LDR. (8)

3.3 An ultrasonic sensor receives its echo 0,004 s after sending a pulse. The speed of sound is 340 m/s. Calculate the distance to the object. (3)

3.4 Compare a DC motor and a servo motor. Give TWO differences. (4)

3.5 Explain how gears can make a robot slower but stronger. (2)

[20]

QUESTION 4: MOVEMENT AND CONFIGURATIONS

4.1 Explain what is meant by a degree of freedom (DOF). (2)

4.2 Name the FOUR common configurations of robot arms and describe the movement of ONE of them. (5)

4.3 A robot's wheels have a diameter of 60 mm and turn at 100 r/min.

4.3.1 Calculate the speed of the robot in m/s. (3)

4.3.2 Calculate how far the robot travels in 10 seconds. (1)

4.4 Explain how a robot with differential steering turns left and right. (3)

4.5 Explain why a tracked robot is better than a wheeled robot on soft sand. (2)

[16]

QUESTION 5: SAFETY, ETHICS AND MAINTENANCE

5.1 Give FOUR safety rules for building and testing robots in a workshop. (4)

5.2 Give TWO safety precautions when using and charging lithium batteries. (2)

5.3 Explain TWO ethical issues about robots in society. (4)

5.4 Give ONE positive and ONE negative effect that robots may have on jobs in South Africa. (2)

5.5 List FOUR maintenance checks you would do on a mobile robot. (4)

5.6 Explain why cybersecurity is important for robots that are connected to the internet. (2)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER